Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security, Semester - II

Term Assessment – I (February- 2025)

Subject Code: CTMSCS SII P4 **Date:** 13.02.2025

Subject Name: Incident Response and Digital Forensics

Time Duration: 45 minutes **Total Marks: 25**

21/25

Instructions:

All questions are compulsory

Short Answer Questions

1x10 = 10

Q.1 Who first noticed the incident?

Q.2 How does confidentiality differ from integrity in Computer Security?

Q.3 What are the common types of security incidents that require reporting?

Q.4 The role of an incident response team includes containment, eradication, and _____ of incidents..

10/10

Q.5 How was the incident discovered?

Q.6 An attack designed to overwhelm a network or system with traffic, making it unavailable to users, is known as a _____.

Q.7 What preventive measures can be taken to avoid similar incidents in the future?

Q.8 What are common signs of a computer security incident?

Q.9 What are vulnerability resources?

Q.10 The three key concepts of information security are _____, _____, and _____.

Long Answer Questions

Attempt any three questions: 5x3 =15

Q.1 How do malware incidents differ from denial-of-service (DoS) incidents?. 4

Q.2 How does incident response help in minimizing damage during a security breach? 4

Q.3 How can organizations ensure proper documentation of incidents for future learning and compliance?

Q.4 What are the type of computer security incident? Explain in details. 3

Write more.

Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security, Semester - II

Term Assessment – I (February- 2025)

Subject Code: CTMSCS SII P2 **Date:** 11.02.2025

Subject Name: Malware Analysis

Time Duration: 45 minutes **Total Marks: 25**

21/25

Instructions:

All questions are compulsory

Short Answer Questions

1x10 = 10

Q.1 What are the key objectives of malware analysis?

9/10 Q.2 What is malware analysis, and why is it important in cybersecurity?

Q.3 What is the difference between a rootkit and a trojan?

Q.4 Worms differ from viruses because they can _____ without human intervention.

Q.5 How does ransomware typically spread?

Q.6 The process of isolating and running malware in a safe environment is called _____.

Q.7 What is the purpose of sandboxing in malware analysis?

Q.8 How can network traffic analysis help in detecting malware?

Q.9 Static malware analysis examines the malware code without _____ it?

Q.10 Process Injection is a technique used by malware to insert malicious code into legitimate processes.

Long Answer Questions

Attempt any three questions: 5x3 =15

4 Q.1 Compare and contrast static and dynamic malware analysis. When should each method be used?

4 Q.2 Explain the role of sandbox and monitoring a malware in malware analysis.

4 Q.3 What steps should be taken when analyzing a suspicious file in a controlled environment?

Q.4 What is polymorphic and metamorphic malware? How do they evade traditional detection methods?

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security

Term Assessment – I February- 2025

Subject Code: CTMSCS SII P3 **Date:** 12.02.2025

Subject Name: Mobile Security

Time: 10:45 AM **Total Marks:** 25

24/25

Instructions: All questions are compulsory

Short Answer Questions

(1x10=10)

Q.1. What are run time permissions in android?

Q.2. What is application sandboxing in android?

Q.3. Mention at least 4 key features of android.

Q.4. What is ahead of time compiler?

Q.5. Name 2 languages used to develop android applications.

Q.6. What is the function of manifest.xml file?

Q.7. What is Dalvik?

9.5/10

Q.8. What is a service in android?

Q.9. Mention important partitions in android.

Q.10. What are signature permissions in android?

Long Answer Questions

Attempt any three questions: (5x3 =15)

Q.1. How does inter process communication happen in android?

Q.2. Explain android's architecture in detail. 5

Q.3. Explain android boot process in detail. 5

Q.4. Explain lifecycle of an activity in android. 4.5

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY, BHOPAL

M.Sc. Cyber Security

Term Assessment – I February- 2025

Subject Code: CTMSCS SII P1 **Date:** 10.02.2025

Subject Name: Network Security

Time: 10:45 AM **Total Marks:** 25

19/25

Instructions: All questions are compulsory

Short Answer Questions

(1x10=10)

Q.1. Define threat and vulnerability.

Q.2. What is the function of DHCP?

Q.3. Name 2 application servers. 1/2

Q.4. Mention the steps in DNS look up.

Q.5. What is the function of datalink layer?

Q.6. What is static and dynamic Ip allocation? 1/2

Q.7. What is ARP poisoning?

Q.8. What is DOS attack?

Q.9. What is a proxy server?

Q.10. Mentions functions of web server.

9/10

Long Answer Questions

Attempt any three questions: **(5x3 =15)**

Q.1. Explain OSI model in detail. 5

Q.2. What are the different types of firewalls?

Q.3. Differentiate between NOC and SOC. 2.5

Q.4. What is IPS? Explain its functions. How is it different from IDS? 2.5

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security & Digital Forensics

Mid Semester Examination, Semester - II - March -2025

Subject Code: CTMSCS SII P4 **Date:20/03/2025**
Subject Name: Incident Response and Digital Forensics
Time: 11:00 AM TO 12:30 PM **Total Marks: 50**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

Q.1.	Answer the following / Define the following (a) What is difference between Events and Incidents? (b) What are the key elements of a nation's cyber strategy? (c) In what scenarios would you choose to use Process Hacker over other system monitoring tools? (d) Justify why continuous monitoring is essential for identifying cybersecurity incidents early. (e) What are the different methods to back up the Windows Registry? (f) Justify how user behavior analytics (UBA) can help identify insider threats and potential security breaches. (g) How does incident response help in minimizing damage during a security breach? (h) What are the types of computer security incident? (i) Why is it important to manage user privileges carefully in both local and network environments? (j) Explain how you would determine if the workstation has been compromised?	10*2 (Marks)=20
Q.2.	Answer the following question in short (any 4) (a) How does an IRT prioritize and classify security incidents in a Windows environment. (b) How does a Threat Analyst assist in identifying and analyzing cyber threats in a Windows environment? (c) What responsibilities does a Forensic Investigator have in analyzing malicious activity on a compromised Windows system? (d) Describe your process to quickly assess the scope of the incident. (e) What tools and techniques do an IRT use for forensic analysis of a compromised Windows system?	4*5 (Marks)=20
Q.3.	Answer the following in detail If the incident is confirmed as a malware infection, what steps would you take to eradicate the threat from the workstation and the network? Explain in detail. OR Identify the most critical immediate actions you would take in the first 15 minutes, and explain your rationale.?	1*10 (Marks)=10

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

School of Cyber Security and Digital Forensics

M.Sc. Cyber Security-Semester-II

Mid Semester Examination – March-2025

Subject Code: CTMSCS SII P3 **Date:19-03-2025**

Subject Name: Mobile Security

Time: 1 hour 30 min **Total Marks: 50**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable diagrams wherever necessary.**
4. **Figures to the right indicate full marks.**

Q.I. Answer the following **10*2 =20 Marks**

1. Write a general outline on mobile pen testing strategy.
2. What are the different types of permission in android?
3. What is the purpose of intents in android?
4. What is an activity in android?
5. Write a short note on android partitions.
6. What are the android application components?
7. How can you install an .apk file in a device using adb?
8. What is improper credential usage vulnerability in android?
9. What is the purpose of APK tool and JD GUI tool?
10. How are android applications compiled and executed?

Q. II. Answer the following questions in short (any 4) **5*4= 20 Marks**

1. What do you understand by secure source code review?
2. How can you perform reverse engineering of an android application? Explain with any tool.
3. What is insecure login vulnerability in android application? Explain with an example.
4. Explain the process you will use to check the logs of android application using adb.
5. Explain insecure data storage issue in mobile application with an example.

Q.III. Answer the following in detail (any 1) **1*10=10 Marks**

1. Explain android boot process in detail.
2. Explain any 5 of the OWASP top 10 vulnerabilities for mobile.

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security & Digital Forensics

Mid Semester Examination, Semester - II - March -2025

Subject Code: CTMSCS SII P2 **Date:**18/03/2025
Subject Name: Malware Analysis
Time: 11:00 AM TO 12:30 PM **Total Marks:** 50

Instructions:
1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

Q.1.	Answer the following / Define the following (a) How does a virus differ from a worm? (b) Explain the function of an accumulator register? (c) How does email work? (d) What is the difference between active and passive cyber-attacks? (e) How can you extract and analyze Windows registry hives from memory using Volatility? (f) What is the purpose of sandboxing in malware analysis? (g) What is a social engineering attack? (h) What is the role of the program counter (PC) register? (i) Why is it important to manage user privileges carefully in both local and network environments? (j) How do you identify whether a file is malicious?	10*2 (Marks)=20
Q.2.	Answer the following question in short (any 4) (a) A forensic investigator suspects that a system was compromised. How can Volatility help in detecting signs of intrusion? (b) What is the Windows Registry, how does it help in detecting malware. (c) What is the difference between hashing and encryption? (d) How does Process Hacker display and organize running processes compared to Windows Task Manager? (e) Explain in detail what is polymorphic and metamorphic malware.	4*5 (Marks)=20
Q.3.	Answer the following in detail How can you extract network connections and registry artifacts from a memory dump using Volatility, and how can this analysis help in detecting malware infections in live memory? OR What are the key techniques used to analyze a malicious Windows program, and how do static and dynamic analysis complement each other in malware investigation?	1*10 (Marks)=10

static
dynamic
hashing
string extraction
identifier rule

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

School of Cyber Security and Digital Forensics

M.Sc. Cyber Security-Semester-II

Mid Semester Examination – March-2025

Subject Code: CTMSCS SII P1 **Date:17-03-2025**
Subject Name: Network Security
Time: 1 hour 30 min **Total Marks: 50**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable diagrams wherever necessary.**
4. **Figures to the right indicate full marks.**

Q.I. Answer the following **10*2 =20 Marks**

1. What are the principles of security?
2. What is promiscuous mode in network?
3. What is affine cipher?
4. Encrypt **Security** using Vignere Cipher using key as **TREE**.
5. Explain transposition cipher with an example,
6. Explain functions of switch and router in a network.
7. What is MAC flooding?
8. How does DHCP protocol work?
9. Encrypt **Forensics** using Caesar cipher with **key = 7**.
10. What is the use of SIEM in a network?

Q. II. Answer the following questions in short (any 4) **5*4= 20 Marks**

1. Explain ARP poisoning with proper diagram.
2. Explain evolution of WEP, WPA and WPA 2 in detail.
3. What is message authentication code? How is it different from hashing?
4. Explain working of digital signatures.
5. Differentiate between symmetric and asymmetric encryption.

Q.III. Answer the following in detail (any 1) **1*10=10 Marks**

1. Explain different types of cipher attacks with an example.
2. What is 802.11 protocols? Elaborate its evolution in detail with a comparison of each protocol.

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination

M. Sc. Cyber Security - Semester – II – April - 2025

Subject Code: CTMSCS SII P4 **Date:** 29/04/2025

Subject Name: Incident Response & Digital Forensics

Time: 10:30 AM to 01:30 PM **Total Marks:** 100

Instructions:

1. **Write down each question on separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Answer the following (Attempt Any Three)**	
(a)	i) Define Incident Response & its need in the domain of cyber security. ii) What are the goals of Incident Response?	08
(b)	Your company has recently hired few interns to help you out in your IRM team. Give a brief idea to the team about the signs of an incident with proper examples.	08
(c)	Explain the categories of Incident Response.	08
(d)	Your department head has called an urgent meeting to discuss the security of client information and you being a security consultant he has asked you to explain the different data classification techniques to the team.	08
Q.2	**Answer the following (Attempt Any Three)**	
(a)	As the cybersecurity lead for a healthcare facility undergoing a ransomware attack that compromises patient records, outline your strategy for incident prioritization based on informational impact. Additionally, provide an example for each category of impact.	08
(b)	How virtualization affects the phases of incident handling?	08
(c)	How can an organization know how much would an incident cost to them? Explain with an example.	08
(d)	Describe the various roles within an incident response team along with their primary responsibilities.	08
Q.3	**Answer the following (Attempt Any Three)**	
(a)	What are the main sources of logs used in Incident Response Management? Briefly describe at least three different log sources and explain how each one aids in understanding and addressing security incidents within an organization's network.	08
(b)	Explain preparation, identification and eradication phases of Incident handling.	08
(c)	Prepare an Incident handler checklist taking a case scenario.	08
(d)	Discuss the phases of containment, eradication & recovery in context to Incident Response Management.	08

Q.4 **Answer the following (Attempt Any Two)**

(a) Explain Digital Forensics and its branches. Also state the importance of Locard's Principle of Exchange in Digital Forensics. 07

(b) What is Chain of Custody and explain its necessity in Digital Forensics? Also state how is integrity maintained in Digital Forensics. 07

(c) Answer the following : 07

i) Write Blockers

ii) Imaging v/s Cloning

Q.5 **Answer the following (Attempt Any Two)**

(a) Explain NTFS & FAT32 File Systems. 07

(b) What are Registry Artefacts and how are they important in establishing a case in Digital Forensics? Explain with examples and artefacts. 07

(c) Consider the following Sysmon Event Log Entry and answer the asked question: 07


```
{
 "Event": {
  "Timestamp": "2023-05-20T13:45:55.000Z",
  "EventID": 1,
  "Provider": "Microsoft-Windows-Sysmon",
  "Channel": "Microsoft-Windows-Sysmon/Operational",
  "Computer": "SERVER-EXAMPLE",
  "EventData": {
   "Image": "C:\\Windows\\System32\\rundll32.exe",
   "ParentImage": "C:\\Windows\\System32\\svchost.exe",
   "ParentCommandLine": "svchost.exe -k netsvcs",
   "TargetImage": "C:\\ProgramData\\Malicious\\malware.exe",
   "TargetCommandLine": "malware.exe -start -mode stealth",
   "GrantedAccess": "0x1410",
   "LogonId": "0x3e7",
   "UtcTime": "2023-05-20 13:45:54.500"
  }
 }
}
```


Explain the following event log parameters & their significance and also Identify the potential threat through the provided event log.

--- End of Paper---

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)
M.Sc. Cyber Security Semester – II

Subject Code: CTMSCS SII P2 **Date: 28/04/2025**
Subject Name: Malware Analysis
Time: 10.30 hrs. – 13.30 hrs. **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

		Marks
Q.1	**Attempt any three.**	
(a)	Discuss process explorer and process monitor tools.	08
(b)	i) What is unpacking stub?	08
	ii) What is keylogger? How it functions?	
(c)	Explain any two encoding techniques with example.	08
(d)	Convert the following disassembly into the C construct code. Calculate the last value of eax, ebx, ecx and edx registers from following disassembly.	08

```
00401006    mov [ebp+var_4], 0Ah
0040100D    mov [ebp+var_8], 9
00401014    mov eax, [ebp+var_4]
00401017    add eax, 0Bh
0040101A    mov [ebp+var_4], eax
0040101D    mov ecx, [ebp+var_4]
00401020    sub ecx, [ebp+var_8]
00401023    mov [ebp+var_4], ecx
00401026    mov edx, [ebp+var_4]
00401029    sub edx, 1
0040102C    mov [ebp+var_4], edx
0040102F    mov eax, [ebp+var_8]
00401032    add eax, 1
00401035    mov [ebp+var_8], eax
00401038    mov eax, [ebp+var_4]
0040103B    cdq
0040103C    mov ecx, 3
00401041    div ecx
00401043    mov [ebp+var_8], edx
```

Q.2	**Attempt any three.**	
(a)	A malware is using a technique which injects code into another running process, and that process executes the malicious code. What kind of technique is this? Explain that in detail with example.	08
(b)	Write a C program for calculating the simple interest, create its assembly code and explain.	08
(c)	Write a detailed note on types of malwares with its examples.	08

(d) Write a detailed and real-world case study of malware attack with all the technical information. 08

Q.3 **Attempt any three.**

(a) Discuss any 08 windows functions/APIs with its malicious use case. 08

(b) Explain the basic static analysis process with a complete example. 08

(c) Malware author uses some techniques to deviate the analyst and makes its analysis difficult for them. Which technique(s) the malware author use here to perform the said task? Explain that/those in detail. 08

(d) Discuss the process of reverse engineering of APK file, also explain the possible suspicious artefacts from its reverse engineering. 08

Q.4 **Attempt any two.**

(a) Write a detailed note on Ollydbg and its features for advance dynamic analysis. 07

(b) Write a short note on anti-malware signature and explain the practical of clam-av to generate the anti-malware signature. 07

(c) Write a detailed note on IDA Pro tool with its functions for advance static analysis. 07

Q.5 **Attempt any two.**

(a) What is the memory of CPU except cache, known as? Explain that memory in detail. 07

(b) Explain the fake net and its significance. How the fake-net can be created practically? 07

(c) Discuss the android architecture in detail. 07

--- End of Paper---

Seat No.: _____ Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (April – 2025)
M.Sc. Cyber Security (Batch 2024-2026)
Semester – II

Subject Code: CTMSCS SII P3 **Date:** 25/04/2025
Subject Name: Mobile Security
Time: 10:30 AM to 01:30 PM **Total Marks:** 100

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Figures to the right indicate full marks.**

			Marks
Q.1		**Answer the following question (Attempt any three)**	
	(a)	Describe Android Architecture in detail with an appropriate diagram.	**08**
	(b)	Describe Android Application Components in detail.	**08**
	(c)	What is Sandboxing? Describe Inter-process Communication in the Android OS.	**08**
	(d)	Explain the Android Boot process in detail.	**08**
Q.2		**Answer the following question (Attempt any three)**	
	(a)	What is ADB? Explain any five ADB commands.	**08**
	(b)	Discuss access control issues and how you would pen-test it.	**08**
	(c)	What is Content Provide Leakage, and how do you pent-test it?	**08**
	(d)	Discuss the client-side injection with an example.	**08**
Q.3		**Answer the following question (Attempt any three.)**	
	(a)	Discuss the Mobile application security pen-testing strategy.	**08**
	(b)	Discuss any four Drozer modules with examples.	**08**
	(c)	Discuss the Static Analysis using MobSF.	**08**
	(d)	Discuss the insecure data storage issue with an example.	**08**
Q.4		**Answer the following question (Attempt any two)**	
	(a)	Discuss how Frida and the Objection Framework will be used to pen-test mobile application security.	**07**
	(b)	Discuss reverse engineering techniques to reverse engineer an Android application.	**07**
	(c)	Discuss the Dynamic Analysis using MobSF.	**07**
Q.5		**Answer the following question (Attempt any two)**	
	(a)	What is the importance of Network Traffic Analysis of an Android Device? How does it help us in penetration testing? Explain.	**07**
	(b)	Describe the difference between Active and Passive network traffic analysis of an Android Device in detail.	**07**
	(c)	Describe the Android Traffic Interception. How can you intercept the HTTP/HTTPS traffic using a Proxy Server? Explain in detail.	**07**

--- End of Paper---

Seat No.: _____ Enrolment No. _____

NATIONAL FORENSIC SCIENCES UNIVERSITY

Semester End Examination (APRIL – 2025)

M.Sc. Cyber Security- Semester - II

Subject Code: CTMSCS SII P1 **Date: 24/04/2025**

Subject Name: Network Security

Time: 10:30 AM to 01:30 PM **Total Marks: 100**

Instructions:

1. **Write down each question on a separate page.**
2. **Attempt all questions.**
3. **Make suitable assumptions wherever necessary.**
4. **Use of Scientific Calculator is allowed.**

			Marks
Q.1		**Attempt any three.**	
	(a)	Compare the ISO/OSI model with the TCP/IP model. Highlight the similarities and differences between the layers of both models and describe how each OSI layer maps to the respective layers in the TCP/IP model.	08
	(b)	What is the CIA triad in network security? Describe each of its components and support your explanation with a practical example for Confidentiality, Integrity, and Availability.	08
	(c)	What is MAC flooding? Discuss the sequence of actions an attacker takes to perform a MAC flooding attack on a network switch.	08
	(d)	An organization wants to secure its network from internal and external threats. Explain how implementing IDS and IPS can help achieve this goal.	08
Q.2		**Attempt any three.**	
	(a)	Outline the complete process of a penetration test, from planning and scoping to reporting the findings.	08
	(b)	Describe the working mechanism of any symmetric encryption algorithm in detail.	08
	(c)	Assume you are setting up a basic RSA encryption system using the primes p=19 and q=23, and a public exponent e=13. Complete the following tasks: i) Determine the public and private keys. ii) Encrypt the message M=3 using RSA encryption.	08
	(d)	Explain the difference between stream cipher and block cipher in detail.	08
Q.3		**Attempt any three.**	
	(a)	Explain the concept of a Digital Signature, and provide a detailed explanation of how the MD5 hash function works.	08
	(b)	Define the terms "attack surface" and "attack vector." Explain each in detail, highlighting their significance in cybersecurity. Additionally, discuss common types of attacks associated with each and suggest countermeasures.	08
	(c)	Explain the methods used by war driving and dumpster diving to collect information. How can businesses protect themselves from these tactics?	08
	(d)	Explain OSCAR Methodology with examples.	08

Q.4		**Attempt any two.**	
	(a)	Provide a detailed explanation of the Transport Layer Security (TLS) protocol, its purpose, and how it works. Also, explain the role of SSL in HTTPS and how it ensures a secure connection between a client and server.	07
	(b)	Explain 802.11 protocol and also discuss the differences between WEP and WPA.	07
	(c)	What is the role of an Access Point (AP) in a WLAN? How does it facilitate communication between wireless and wired devices?	07
Q.5		**Attempt any two.**	
	(a)	What is live forensics, and how does it differ from traditional digital forensics? Explain the key steps involved in conducting a live forensic investigation, and provide examples of the types of evidence that can be gathered in a live forensics scenario.	07
	(b)	Imagine a business suspects unauthorized access to its critical systems. As a digital forensic investigator, explain the steps you would follow in a network forensic investigation. i. What network-based digital evidence would you focus on first? ii. Describe the process of gathering evidence in a network forensic investigation. iii. What challenges do you face in ensuring the admissibility and integrity of digital evidence?	07
	(c)	Define the following terms and explain their roles in network management (Any three): i. DNS Server ii. DHCP Server iii. Proxy Server iv. SOC	07

--- End of Paper---

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security and Digital Forensics

End Sem Practical Examination, Semester - II - May -2025

Subject Code: CTMSCS SII L4 **Date:08/05/2025**
Subject Name: Incident Response and Digital Forensics Laboratory
Time: 3 hours **Total Marks: 100**

Instructions: You can use any malware of your choice.

I. Attempt all questions

Q1. Perform demonstration for the process of real-time log capture, event monitoring, and basic log analysis to detect suspicious activities using Sysmon and Event Viewer. **1*15=15 marks**

Q2: Identify how programs are configured to automatically start on system boot using Registry keys and Analyze registry hives to extract user activity and configuration history. **1*20=20marks**

Q3: To simulate a basic malware infection scenario and analyze the system for signs of compromise. **1*15=15 mark**

III. Record **20 marks**

IV. Viva **30 marks**

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security and Digital Forensics

End Sem Practical Examination, Semester - II - May -2025

Subject Code: CTMSCS SII L3 **Date:07/05/2024**
Subject Name: Mobile Security Laboratory
Time: 3 hours **Total Marks: 100**

Instructions:
1. **Attempt all questions**

I. Perform the following:

1. You are given the apk file of a mobile application InsecureShop. Perform any 3 OWASP top 10 vulnerabilities on the application. **(30 marks)**
2. Use any reverse engineering tool to locate the vulnerability inside the application. **(10 marks)**
3. Prepare a security audit report of the application. **(10 marks)**

III. Record **20 marks**

IV. Viva **30 marks**

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security and Digital Forensics

End Sem Practical Examination, Semester - II - May -2025

Subject Code: CTMSCS SII L2
Subject Name: Malware Analysis Laboratory **Date:06/05/2025**
Time: 3 hours

Total Marks: 100

Instructions: You can use any malware of your choice.

I. Attempt all questions

Q1. You are provided with a sample of a suspicious executable file. Use a sandbox environment to run the file and observe its behavior.
Tasks: **1*20=20 marks**

1) List any files created or modified by the malware.
2) Identify any processes spawned.
3) Note any network connections attempted.

Tools Recommended: Cuckoo Sandbox, Procmon, Process Explorer, Wireshark

Q2: Analyze the malware's network behavior.
Tasks: **1*10=10 marks**

1) Capture and analyze network traffic generated during execution.
2) Identify any IP addresses or domain names contacted.
3) Determine if the malware attempts data exfiltration or not.

Tools Recommended: Wireshark, TCPView, FakeNet-NG

Q3: Run the malware in a controlled environment and monitor any registry or file system changes. **1*20=20 marks**
Tasks:

- Document any registry keys created or modified.
- Record file paths, types, and timestamps.
- Describe the significance of these changes (e.g., persistence techniques).

Tools Recommended: Regshot, Process Monitor, Autoruns

III. Record **20 marks**

IV. Viva **30 marks**

Enrolment No._____

NATIONAL FORENSIC SCIENCES UNIVERSITY
School of Cyber Security and Digital Forensics

End Sem Practical Examination, Semester - II - May -2025

Subject Code: CTMSCS SII L1 **Date:05/05/2024**
Subject Name: Network Security Laboratory
Time: 3 hours **Total Marks: 100**

Instructions:
1. **Attempt all questions**

I. Perform the following:

1. Use cisco packet tracer to simulate routing using switch and router in 2 LANS. **(20 marks)**
2. Perform network fingerprinting using nmap or similar tools to identify operating systems, open ports, and services running on remote hosts within a simulated or controlled network environment. **(15 marks)**
3. Demonstrate the use of wireshark to capture, filter, and analyze network traffic in order to identify protocols, extract useful data, and detect potential security threats. **(15 marks)**

III. Record **20 marks**

IV. Viva **30 marks**